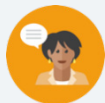


2.3 Cube Values

Lesson Introduction

 Benchmark	MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$, where p is a whole number and q is an integer, determine the real solutions.		 Learning Targets	- I can use multiple representations to explore cube values. - I can determine the positive and negative solutions for $x^3 = q$, where q is an integer. - I can find the values of cube roots.
 Benchmark Coherence	Building On		Working Toward	
	N/A		MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real-number system.	
 Essential Questions	- What does it mean to cube a number? - What is a perfect cube? - How can perfect cubes be represented geometrically? - How can perfect cubes be represented algebraically?			
 Lesson Narrative	This is the first of two lessons in which students learn about perfect cubes and cube roots as they apply their understanding of perfect squares and square roots to explore perfect cubes and cube roots. Students connect the geometric representation of the volume of a cube to the algebraic equation $x^3 = q$. Students graph the coordinate pairs to discover that perfect cubes can be positive or negative. The conceptual foundation that the lesson lays should not be rushed. Allow students to explore and ask questions.			
 Component Summary	2.3.1 Warm-Up (~5 min.)	Solve a puzzle (SE p. 74)	2.3.4 Exploration (10 min.)	Explore the graph of $x^3 = y$ and values that satisfy this equation (SE p. 77)
	2.3.2 Exploration (10 min.)	Relate the side lengths and volumes of cubes (SE p. 75)	2.3.5 Guided Instruction (10-15 min.)	Learn to solve equations in the form $x^3 = q$ (SE p. 78)
	2.3.3 Bring It Together (5 min.)	Discuss and summarize similarities and differences between square and cube numbers (SE p. 76)	2.3.6 Wrap-Up (~5 min.)	Solve equations in the form $x^3 = q$ and evaluate cube roots (SE p. 79)
 Resources	Glossary		Materials	
	<u>Key Terms</u> : N/A <u>Additional Terms</u> : N/A		(Optional) Unifix cubes (Optional) Chart paper (Optional) Markers	
			Consider allowing students to use a graphing utility for question 1c in 2.3.4 Exploration.	

 Teacher Tips	Common Misconceptions - Students may confuse the exponent for a factor rather than a power. For example, 3^3 might be calculated as 9 rather than 27. - Students may neglect to include the negative sign with each factor when evaluating perfect cubes. For example, they may calculate $-3 \times 3 \times 3$ or $-3 \times -3 \times 3$ instead of $-3 \times -3 \times -3$.	Things to Consider Students explore and create a definition for a perfect cube. A perfect cube is a number that results from multiplying a number by itself three times. Consider building a common definition built around this concept during the discussion at the end of 2.3.2 Exploration.
	Literacy and Language Routines Think-Pair-Share In 2.3.4 Exploration, prompt students to initially complete the table and plot the points individually and then discuss their work with a partner. Finally, debrief the activity with the whole class by asking a variety of students to share their strategies and answers.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2.1: Multiple Representations Students represent cube values both geometrically and algebraically. Students work with the volumes of cubes in 2.3.2 Exploration. Then students complete a table of values and graph the coordinates of the cubic equation, $x^3 = y$, in 2.3.4 Exploration. Discuss the similarities and differences between these representations to build students' conceptual understanding.
	 Differentiate Instruction	Consider providing a kinesthetic entry point by having students model the volumes of cubes using Unifix cubes in 2.3.2 Exploration. A visual entry point can also be provided by using a virtual manipulative of Unifix cubes, and Desmos can be used to graph $x^3 = y$ in 2.3.4 Exploration. Additional differentiation opportunities are denoted throughout the lesson guide.
Lesson Notes:		

2.3.1 Warm-Up: Magic Square

Component Overview: Students use the digits one through nine to complete a magic square where the sum of each row, column, and diagonal is 15.



2.3.1 Warm-Up: Magic Square

- Place the digits 1 to 9 in each square so the sum of the digits in each row, column, and diagonal is 15. Each digit can only be used once.

Answers will vary

4	9	2
3	5	7
8	1	6



If students enjoyed the Magic Square, consider assigning students to challenge their other teachers, peers, or family members to their own Magic Square. Prompt students to find their own sum or product, verify their solutions, and describe who they assigned it to and who completed it correctly. Give points for accuracy and creativity.

2.3.2 Exploration: Volume of Cubes

Component Overview: Students explore the relationship between the side lengths and volumes of cubes as they develop a sense of what it means for a number to be a perfect cube.



2.3.2 Exploration: Volume of Cubes

The volume of a cube can be found by multiplying the length by the width by the height. Volume is measured in cubic units.

Work with your partner to complete the following.

1. Complete the statements.

The length, width, and height of a cube have

- ☒ the same
- ☐ different

measure(s). Therefore, the volume of a cube can be found

by taking one side length to the

- ☐ first
- ☐ second
- ☒ third

power, which is

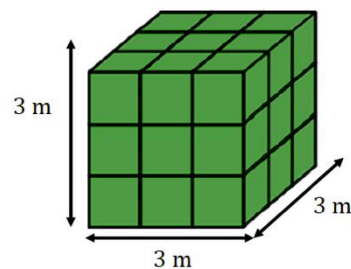
also known as

- ☐ squaring
- ☒ cubing

the side length.

2. A cube is shown with dimensions given in meters (m).

Complete each statement.



- a. The edges of the cube each measure 3 m.
- b. The volume of the cube is 27 m³ because $3^3 = \underline{27}$.



This Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the Exploration as they engage with each new reflection.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group in 2.3.3 Bring It Together.

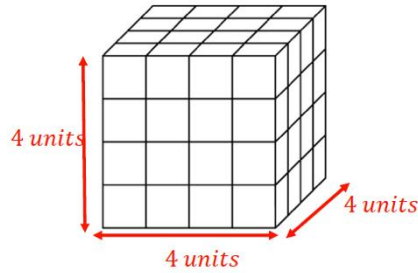
Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.



Consider providing a kinesthetic entry point by having students model the volumes of cubes using Unifix cubes.

2.3.2 Exploration: Volume of Cubes (continued)

3. A cube is shown, where each gridline measures 1 unit.



- Label the measures of the length, width, and height of the cube.
 - What is the volume of the cube?
 $4 \times 4 \times 4 = 64$ units cubed
4. The numbers 27 and 64 are cube numbers, or perfect cubes.
- Explain what you think a perfect cube is.
Answers will vary. I think it is the resulting number when you multiply a number by itself three times. I think it is similar to squaring a number but instead of multiplying the same number by itself twice, you multiply it by itself three times.
 - Work with your partner to find at least two other numbers that are perfect cubes.
Answers will vary. 8; 125; 216; 343; 512



Students do not have to have a precise definition for what it means to be a perfect cube. Students should understand a perfect cube is any value that results from multiplying a number by itself three times. Consider revisiting question 1 to move students toward this understanding.



Ask students to publicly share their responses to question 3b. Compile a variety of responses using the general structure of $(\quad)^3 = __$. Highlight students who cubed negatives, decimals, or fractions so students can see that perfect cubes are not limited to positive whole numbers.

If time allows, consider engaging students in a Notice and Wonder routine based on the patterns they observe (MA.K12.MTR.5.1).

2.3.3 Bring It Together: Perfect Cubes

Component Overview: Students synthesize the similarities and differences between square and cube numbers.



2.3.3 Bring It Together: Perfect Cubes

1. Discuss with your partner the similarities and differences between square numbers and cube numbers. Then, summarize your discussion in the table.

Similarities	Differences
<i>Answers will vary.</i> • Both can be related to geometric figures. • Both are the result of multiplying a number by itself some number of times.	<i>Answers will vary.</i> • Perfect squares are the product when a number is multiplied by itself 2 times, while perfect cubes are the product when a number is multiplied by itself 3 times. • Perfect squares relate to a 2-dimensional figure while perfect cubes relate to a 3-dimensional figure.



Another key difference is that perfect cubes can be negative while perfect squares cannot. This is not addressed here, but students could come back and add this difference to the list after completing 2.3.5 Guided Instruction.

2.3.4 Exploration: Cube Values on the Coordinate Plane

Component Overview: Students explore the equation and graph of $y = x^3$ to develop a deeper understanding of the relationship between a value and its cube.



2.3.4 Exploration: Cube Values on the Coordinate Plane

1. Use the equation $y = x^3$ to complete the following with your partner.
 - a. Complete the table of values for the equation $y = x^3$ by substituting the given values of x and solving for the corresponding y -value.

x	Substitute into $y = x^3$	Coordinates (x, y)
-2	$y = (-2)^3$ $y = -8$	$(-2, -8)$
-1	$y = (-1)^3$ $y = -1$	$(-1, -1)$
0	$y = 0^3$ $y = 0$	$(0, 0)$
1	$y = 1^3$ $y = 1$	$(1, 1)$
2	$y = 2^3$ $y = 8$	$(2, 8)$

- b. The y -values are cube numbers. What do you notice about the y -values?

Answers will vary. I noticed that unlike when you square a number, some of the y -values are negative. I noticed that when you cube a positive number, the answer is positive, but when you cube a negative number the answer is negative.

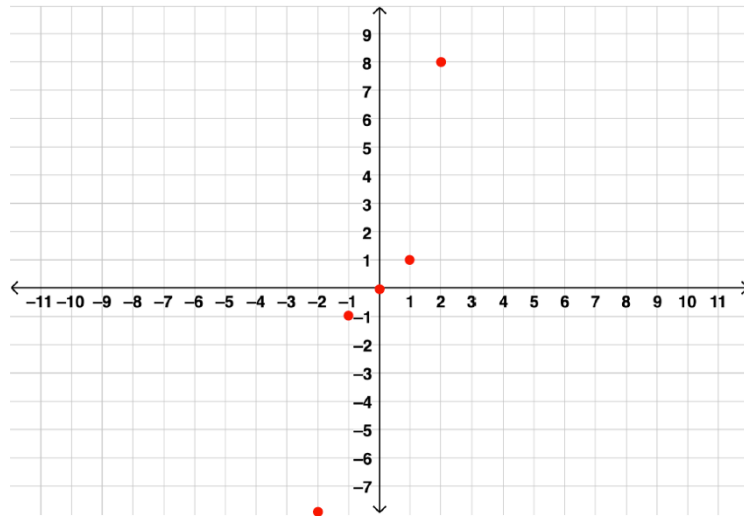


The goal of this Exploration is to show students another representation of perfect cubes. The geometric representation explored in 2.3.2 Exploration could lead to a misconception that perfect cubes are always positive. Discuss that the perfect cubes in 2.3.2 Exploration were all positive because lengths are always positive. Prompt students to compare and draw connections among these representations.

Exploring the graph of this relationship allows students to better understand this relationship while also reinforcing the relationships between equations, tables of values, and graphs, which will be reinforced in future units throughout the course.

2.3.4 Exploration: Cube Values on the Coordinate Plane (continued)

c. Plot the coordinates on the grid shown.



Notice the graph of $y = x^3$ includes points above and below the x -axis, unlike the graph of $y = x^2$ explored in a previous lesson.



What do you notice about the relationship between the x – and y –values? How does this compare to the coordinate pairs of $y = x^2$?



Students do not actually need to draw the curve, as this goes beyond what they need to understand here. However, consider showing them the graph of $y = x^3$ in Desmos so that they understand what it looks like.



Challenge students to substitute fractions and decimals into the equation $y = x^3$ and plot the coordinates on the graph.

2.3.5 Guided Instruction: Solving Equations in the Form $x^3 = q$

Component Overview: Students learn to solve equations in the form $x^3 = q$, where q is an integer.



2.3.5 Guided Instruction: Solving Equations in the Form $x^3 = q$

1. Find the solution to each equation.

a. $x^3 = 8$

2

b. $-8 = x^3$

-2

2. Discuss with your partner the similarities and differences between the equations and their solutions.

Answers will vary. They are similar because we think about what number can be multiplied by itself three times to solve them both. They are different because each equation only has one solution, either positive or negative.

3. Complete the statements.

For equations in the form $x^3 = q$, where q is an integer,

there ☐ is only one solution.
☐ are two solutions.
☐ are three solutions.

The solution(s) can

☐ only be positive.
☐ only be negative.
☐ be positive or negative.

The sign of the solution(s)

depends on the sign of q .



Model how to solve each equation by modeling through a Think Aloud, asking and answering the following question for each equation.

What number(s), when multiplied by itself three times, results in ____?

Then ask students to consider:

What is the relationship between the sign of x and the sign of q ?



This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference.

Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address before completing question 3.

2.3.5 Guided Instruction: Solving Equations in the Form $x^3 = q$ (continued)

Similar to the relationship between squares and square roots, there is an inverse relationship between cubes and cube roots.

The symbol $\sqrt[3]{}$ is used to indicate a **cube root**, where 3 is in the index of the radical.

Cube roots can be used to solve equations in the form $x^3 = q$, where q is an integer.

4. Complete the table for each equation.

Equation	Related Cube Root	Solution
$x^3 = 27$	$\sqrt[3]{27} = x$	$x = \underline{3}$
$p^3 = -1$	$\sqrt[3]{-1} = p$	$\underline{-1} = p$
$\underline{64} = s^3$	$\sqrt[3]{64} = s$	$s = \underline{4}$
$w^3 = \underline{216}$	$\sqrt[3]{216} = w$	$w = 6$
$\underline{-27} = c^3$	$\sqrt[3]{-27} = c$	$c = \underline{-3}$
$-125 = b^3$	$b = \sqrt[3]{-125}$	$\underline{-5} = b$



Collect and Display

Consider completing the first two rows in the table in question 4 by modeling for students, thinking aloud while filling in the rows. Emphasize the mathematical language that will help students connect cubes and cube roots by noting that they are inverse operations. Invite students to talk through each of the next rows with their partner, recording some of the math talk that they use on chart paper. Ask them to explicitly connect the phrase they used when explaining with the notation in the chart.



After noting that perfect cubes can be negative, encourage students to go back and add this difference to the table in 2.3.3 Bring It Together.

2.3.6 Wrap-Up: Ticket Out the Door

Component Overview: Students evaluate cube roots and solve equations involving perfect cubes. Students should not use a calculator to complete the Wrap-Up. Consider using data from this formative assessment to determine which students need remediation before the next lesson.



2.3.6 Wrap-Up: Ticket Out the Door

- Find the value of each of the following.

a. $\sqrt[3]{125} = 5$

b. $\sqrt[3]{-8} = -2$

- Solve each equation.

a. $w^3 = -64$
 $\sqrt[3]{w} = \sqrt[3]{-64}$
 $w = -4$

b. $1 = k^3$
 $\sqrt[3]{1} = \sqrt[3]{k}$
 $1 = k$



Why is the answer to 1a positive when the answer to 1b is negative? Would this also be true for the fourth root ($\sqrt[4]{x}$)?



Could k also be -1 ? Why or why not?